

Name _____

Algebra I Regents
Practice Test

Grade **8**

ALGEBRA I (COMMON CORE)

The University of the State of New York
REGENTS HIGH SCHOOL EXAMINATION

ALGEBRA I (Common Core)

Student Name: _____

School Name: _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for Part I has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 35 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in Parts II, III, and IV directly in this booklet. All work should be written in pen, except graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice...

A graphing calculator and a straightedge (ruler) must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

ALGEBRA I (COMMON CORE)

Part I

Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet.

- 1 Solve for x in the inequality $3x + 7y > 2x - 8y$.

Use this space for
computations.

- | | |
|--------------|----------------|
| (1) $x > -y$ | (3) $x < -15y$ |
| (2) $x < -y$ | (4) $x > -15y$ |

- 2 If $g(x) = -4x + 5$, what is the value of $g(-3)$?

- | | |
|--------|--------|
| (1) -2 | (3) 17 |
| (2) -7 | (4) 4 |

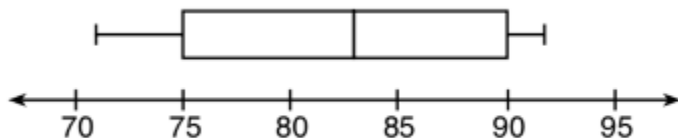
- 3 Which of the following is equivalent to the expression $-3(x^2 + 2x - 3) + 4(4x^2 - 7x + 5)$ when it is written in standard form?

- | | |
|------------------------|-------------------------|
| (1) $13x^2 - 34x + 29$ | (3) $-19x^2 + 22x - 11$ |
| (2) $13x^2 + 22x - 11$ | (4) $19x^2 - 34x + 29$ |

- 4 The zeros of the function $h(x) = x^2 - 5x - 24$ are

- | | |
|--------------|--------------|
| (1) -8 and 3 | (3) -4 and 6 |
| (2) -6 and 4 | (4) -3 and 8 |

- 5 The box plot below summarizes the data for the average monthly high temperatures in degrees Fahrenheit for Nashville, Tennessee.



Use this space for computations.

The third quartile is

- (1) 92 (3) 83
(2) 90 (4) 71
- 6 When written in factored form, $4z^2 + 11z - 3$ is equivalent to

- (1) $(4z + 1)(z - 3)$ (3) $(2z + 1)(z - 3)$
(2) $(2z - 1)(2z + 3)$ (4) $(4z - 1)(z + 3)$

- 7 Students were asked to name their favorite milkshake flavor between chocolate, vanilla, and strawberry. The results are shown in the table below.

	Chocolate	Vanilla	Strawberry
7th Graders	42	58	20
8th Graders	84	41	5

What percentage of the students chose vanilla as their favorite flavor?

- (1) 39.6% (3) 50.4%
(2) 41.4% (4) 58.6%
- 8 Which of the following is *not* a point on the graph of $y = 3x^2 - x + 7$?

- (1) $(-1.5, 15.25)$ (3) $(1.25, 10.25)$
(2) $(0.5, 7.25)$ (4) $(2.5, 23.25)$

Use this space for computations.

- 9** In the balcony of Sharp Theater at the Juilliard School, the number of seats per row increases at a constant rate. Esteban counts 31 seats in row 3, and 37 seats in row 6. How many seats are in row 20?

(1) 65
(2) 67

(3) 69
(4) 71

- 10** Aretha wants to buy snacks for her sleepover with her friends this weekend. Cheetos cost \$1.60 per bag, and Takis cost \$1.75 per bag. If she only has \$10 to spend on snacks, which inequality represents the situation where she buys c bags of Cheetos, and t bags of Takis?

(1) $1.60c + 1.75t \leq 10$
(2) $1.60c + 1.75t \geq 10$

(3) $1.75c + 1.60t \leq 10$
(4) $1.75c + 1.60t \geq 10$

- 11** A function is defined as $\{(0, 1), (2, 3), (a, b), (5, 8), (7, 2)\}$. Isaiah needs to determine values for the ordered pair (a, b) that will keep this set a function. Which of the following ordered pairs could Isaiah pick?

(1) $(0, 2)$
(2) $(5, 3)$

(3) $(7, 0)$
(4) $(1, 3)$

- 12** If the quadratic equation $(x + p)^2 = 16$ has two solutions, $x = 2$ and $x = -6$, what is the value of p ?

(1) -2
(2) -4

(3) 2
(4) 4

- 13** Given the quadratic functions below, which of the following has the largest maximum value over the set of real numbers?

Use this space for computations.

(1) $g(x) = -x^2 + 2x + 4$

(3) $f(x) = -(x - 5)^2 + 5$

(2)

x	h(x)
-2	-9
-1	-3
0	1
1	3
2	3
3	1

(4)

x	k(x)
-1	-1
0	3
1	5
2	5
3	3
4	-1

- 14** Which of the following situations is *not* a linear function?

- (1) A cable television company charges a \$50.00 set-up fee and \$12.00 per month
- (2) A \$5,000 motorcycle depreciates 12% per year
- (3) A taxi company charges \$3.00 originally and \$5.00 per kilometer
- (4) A grocery store cashier earns \$15.00 per hour

- 15** The Saint Silvester Road Race is a famous 15-kilometer race that takes place in Brazil every year. Ezekiel is planning on participating and has done the following training runs:

- I. 10 miles
- II. 15,560 yards
- III. 44,880 feet

Which training run(s) are at least 15 kilometers?

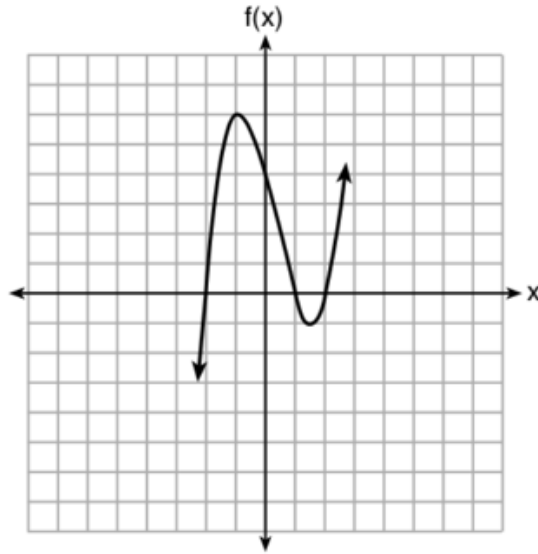
- (1) I only
- (2) II only
- (3) I and III
- (4) II and III

- 16** Which of the following gives the range of the function $h(x) = x^2 + 5$ in interval notation?

Use this space for computations.

- (1) $(-\infty, \infty)$ (3) $[5, \infty)$
(2) $(-\infty, 5]$ (4) $[0, \infty)$

17



Which of the following functions could represent the polynomial function graphed above?

- (1) $f(x) = (x + 1)(x^2 + 2)$ (3) $f(x) = (x - 1)(x^2 - 4)$
(2) $f(x) = (x - 1)(x^2 - 2)$ (4) $f(x) = (x + 1)(x^2 + 4)$

- 18** The amount Ryan earns working at Dunkin Donuts each week can be calculated by the expression $150 + 3.25d$, where d is the number of donuts he sells that week. What is the constant term in this expression, and what does it represent?

- (1) $3.25d$, the amount he earns when he sells d donuts
(2) $3.25d$, the amount he is guaranteed to be paid each week
(3) 150, the amount he is guaranteed to be paid each week
(4) 150, the amount he earns when he sells d donuts

- 19** A homework problem asked scholars to identify which of the following polynomials were written in standard form, and explain how they knew their response was correct.

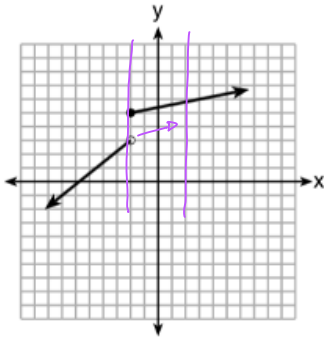
- I. $15x^4 - 6x + 3x^2 - 1$
 II. $12x^3 + 8x + 4$
 III. $2x^5 + 8x^2 + 10x$

Which of the following scholar responses is correct?

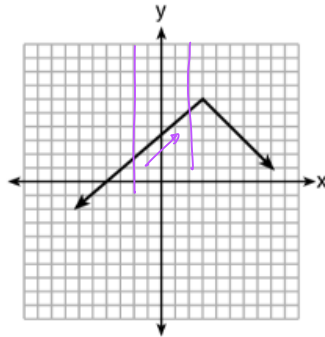
- (1) I and II because the coefficients are decreasing.
 (2) II because the numbers are decreasing.
 (3) II and III because the exponents are decreasing.
 (4) II and III because they each have three terms.

- 20** Which of the following functions is *not* always increasing for all values of x , $-2 < x < 2$?

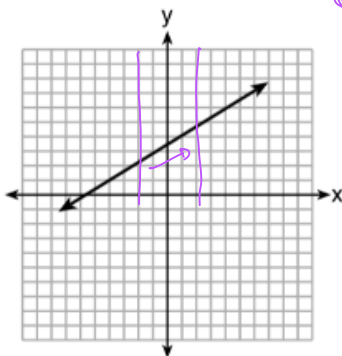
(1)



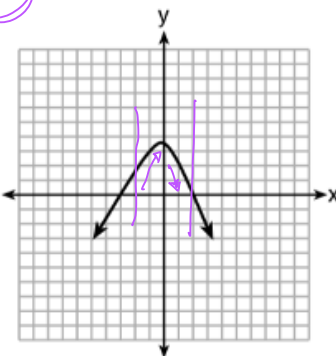
(3)



(2)



(4)



Use this space for computations.

cost $\downarrow$
 $G(w)$ $\rightarrow$ Variable = input = independent = domain
 $\rightarrow w = \text{pounds of grapes}$

Use this space for computations.

- 21 A local supermarket sells bags of grapes. The function $G(w)$ represents the cost, in dollars, of a bag of grapes weighing w pounds. What is the most appropriate domain for this function?

- (1) integers
 (2) rational numbers
 (3) positive integers
 (4) positive rational numbers
- $\rightarrow$ no negative weight

- 22 The following table shows the grades, out of 100, that Jennifer received for thirteen different quizzes in math class.

84	80	87	75	77	79	80	74	76	80	80	82	82
----	----	----	----	----	----	----	----	----	----	----	----	----

The standard deviation of these data is approximately

- (1) 3.5
 (2) 13
 (3) 79.7
 (4) 80

- 23 Washington was asked to write a formula for the length of a rectangle using the formula for its perimeter, $p = 2l + 2w$. Washington was not sure how to answer this question, so he drafted three possible responses.

(I) $l = \frac{1}{2}p - w$
 (II) $l = \frac{1}{2}(p - 2w)$
 (III) $l = \frac{p-2w}{2}$

$p = 2l + 2w$
 $-2w$ $-2w$
 $\hline$
 $p - 2w = 2l$
 $\frac{p-2w}{2} = \frac{2l}{2}$
 $\frac{p-2w}{2} = l$
 $\frac{1}{2}(p-2w)$
 $\frac{p}{2} - \frac{2w}{2} \rightarrow \frac{1}{2}p - w$
 $\frac{1}{2} \& \frac{1}{2}$ reciprocal

Which of Washington's responses are equivalent to the original equation?

- (1) I and II
 (2) II and III
 (3) I and III
 (4) I, II, and III

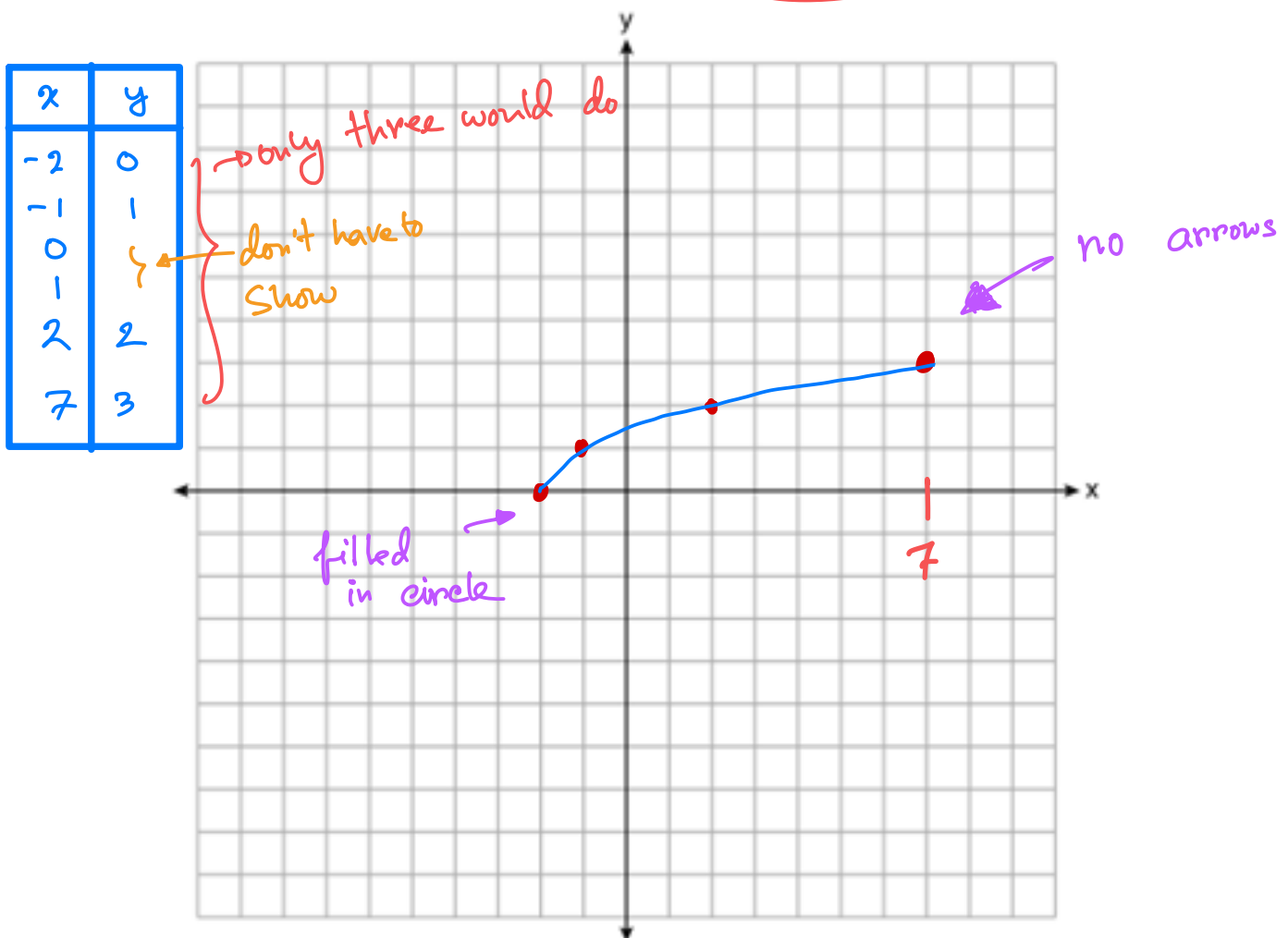
- 24 For a given sequence, if $a_n = n(a_{n-1})$ and $a_1 = 1$, what is the value of the fifth term, a_5 ?

- (1) 5
 (2) 20
 (3) 120
 (4) 720

Part II

Answer all 6 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.

- 25 Graph the function $g(x) = \sqrt{x + 2}$ over the domain $-2 \leq x \leq 7$.



- 26** Princeton states that the function generated from the values in the table shown below is not linear.

x	f(x)
0	2
1	4
2	8
3	16

Handwritten annotations to the right of the table:

- Red arrows pointing from x=0 to x=1, x=1 to x=2, and x=2 to x=3, each labeled $\times 2$.
- Orange curved arrows pointing from f(0)=2 to f(1)=4, f(1)=4 to f(2)=8, and f(2)=8 to f(3)=16, each labeled $\times 2$.

State if Princeton is correct. Explain your reasoning.

27 Solve $6x^2 = 294$ algebraically.

$-294 - 294$

$$\boxed{6x^2 - 294 = 0}$$

QF

$$\frac{6x^2}{6} = \frac{294}{6}$$

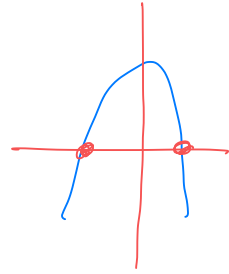
$$\Rightarrow x^2 = 49$$

$$\Rightarrow \sqrt{x^2} = \sqrt{49}$$

$$\Rightarrow \boxed{x = \pm 7}$$

- 28** The function $h(t) = -16t^2 + 256$ models the height of an object, in feet, t seconds after it is dropped from the top of a building. Determine, algebraically, the number of seconds it takes for the object to hit the ground.

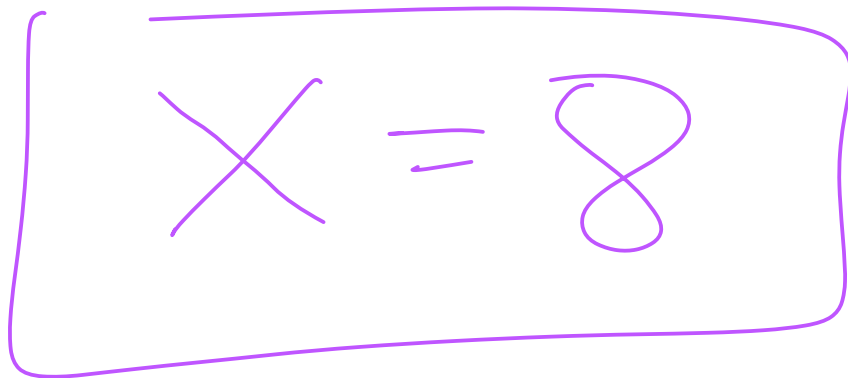
$$t = 4, -4$$



$$t = 4s$$

29 Solve algebraically for x:

$$-\frac{2}{3}(x + 12) + \frac{2}{3}x = -\frac{5}{4}x + 2$$



A handwritten purple box containing the solution $x = 8$.

30 Determine if the product of $\sqrt{9}$ and $\frac{1}{7}$ will be rational or irrational. Explain your reasoning.

$$\sqrt{9} \cdot \frac{1}{7}$$

$$= 3 \cdot \frac{1}{7} = 0.\overline{428571429}$$

$$= \frac{3}{7} \Rightarrow \text{Ratio} \Rightarrow \text{Rational}$$

Part III

Answer all 4 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.

- 31 The population of mice at Rockefeller Research Laboratories can be modeled by the function $m(x) = 30(1.016)^x$. The variable x represents the number of days since the population was first tallied.

Explain what 30 and 1.016 represent in the context of the problem.

$$m(x) = 30(1.016)^x$$

IV ←

↓
Multiplies

1.6% growth rate

Determine the average rate of change from day 40 to day 100 to the nearest tenth.

- 32** There are two ice cream parlors in Bedford Falls. Tasty Treat charges \$7.00 for the first two scoops of ice cream, and \$3.00 for each additional scoop. Nina's Custard charges \$3.25 per scoop of ice cream.

If a customer orders at least three scoops of ice cream, write equations to model the cost of their ice cream with s scoops at both ice cream parlors. Use T for the cost at Tasty Treat and N for the cost at Nina's Custard.

Determine algebraically the number of scoops when the cost of an ice cream will be the same at both locations.

33 Graph the following system of inequalities on the set of axes below.

② $y > -(2-4x)$

or $y > 4x - 2$

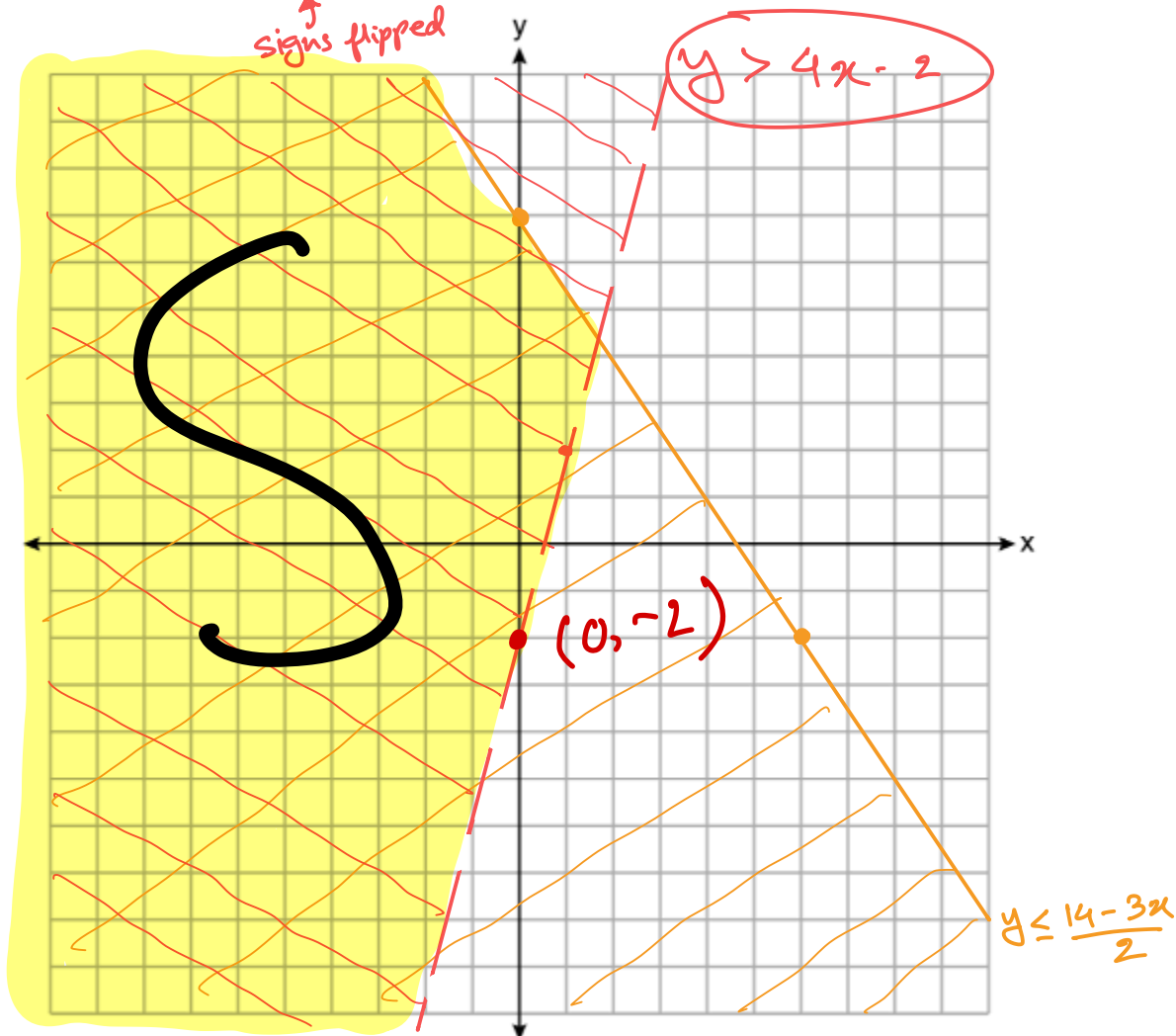
$$\begin{array}{r} 4x - y < 2 \\ -4x \quad -4x \\ \hline -y < 2 - 4x \\ \hline -1 \quad -1 \\ \hline y > -(2 - 4x) \end{array}$$

signs flipped

$$\begin{array}{r} 2y + 3x \leq 14 \\ 4x - y < 2 \end{array}$$

$$\begin{array}{r} 2y + 3x \leq 14 \\ -3x \quad -3x \\ \hline 2y \leq 14 - 3x \\ \hline \frac{2y}{2} \leq \frac{14 - 3x}{2} \\ y \leq \frac{14 - 3x}{2} \end{array}$$

① $y \leq \frac{14 - 3x}{2}$



Determine if the point $(0, -2)$ is in the solution set. Explain your answer.

No, only satisfies inequality ①

- 34 The percentage of scholars scoring 85 or better on a science final and a history final exam during the previous school year for seven Success Academy middle schools is shown in the table below.

Percentage of Students Scoring 85 or Better	
Science, x	History, y
27	46
12	28
13	45
10	34
30	56
45	67
20	42

Write the linear regression equation for these data, rounding all values to the *nearest hundredth*.

①

$$y = \dots +$$

State the correlation coefficient of the linear regression equation, to the *nearest hundredth*.
Explain the meaning of this value in the context of these data.

$$r = 0.92$$

$$r^2 =$$

Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.

35 At Pet Center in Old Bridge, New Jersey, the number of dogs, d , is initially five less than twice the number of hamsters, h . If the store adds three more of each animal, the ratio of hamsters to dogs will be $\frac{3}{4}$.

Write an equation or system of equations that can be used to find the number of hamsters and dogs at Pet Center.

Could Pet Center initially have 15 hamsters and 20 dogs? Explain your reasoning.

Determine algebraically the number of hamsters and dogs that were originally available at Pet Center.

ALGEBRA I PRACTICE TEST